blood Exchanged - 3-4 times/minute Reduced blood pressure in intervillous space helps the Oxygenated blood to the chorionic plate, return back to the villi.

Functions of Placenta

1. **Respiratory:** Placenta act as lungs to the fetus taking in oxygen from the mother's haemoglobin and giving of CO_2 into the maternal blood.
2. **Nutritive:** The fetus selects from the mother blood protein for tissue building, glucose for energy and growth. Calcium and phosphorus for bones and teeth, vitamins, iron and other minerals for blood formation.
3. **Storage:** The liver is not sufficiently developed. Placenta stores glucose is the form of glycogen and reconverts it into glucose as required by the fetus.
4. **Excretory:** The waste products are given off and taken away by the ovarian and uterine veins.
5. **Protective:** To protect the fetus, the placenta prevents a number of organisms from passing through into the fetal blood.
6. **Endocrine:** The placenta also has an endocrine action producing hormones like follicular stimulating and leutinizing hormones of the gonadotrophic hormone and oestrogen and progestetone.

5.6 UMBILICAL CORD (FUNIS)

- **Usually it is attached near the center of the fetal surface of placenta.** The cord extends from the umbilicus of the fetus to the fetal surface of the placenta.
- **Length:** About 40- 50 cm and 20 cm wide.
- Contains two arteries and one vein, surrounded by clear gelatinous substances and gives support to the cord and prevent compression of the cord is called (Wharton jelly).
- Two arteries that bring waste products and deoxygenated blood from the fetus to the placenta and the vein carries oxygenated and nourished blood from the placenta to the fetus. The blood flow through the cord is around 400 ml/mt. At birth when the lungs start to work the function of the placenta ceases.
- The vessels are longer than the cord and may have loops (false knots).

5.7 AMINIOTIC FLUID

- It is a fluid in the uterus surrounding the fetus.
- It consists of fetus urine, maternal serum fetal cells , near birth – amnionic volume will be 500-1000 ml. If the fluid is excess (>2000 ml) it is called as Hydraminos. If less it is called as Oligohydramnios .

Functions of Amniotic Fluid

- It allows for growth and free movement of fetus.
- It protects the fetus, acting as shock absorber.
- It maintains an even temperature for fetus.
- It prevents pressure on the cord.
- It acts as cushion around the fetus, because it protects the fetus from injury, if mother is bumped or falls.
- It is a fluid source that the fetus drinks and then urinates.

XII_Nursing_Vocational_Unit 5.indd 154 1/11/2020 2:46:13 PM